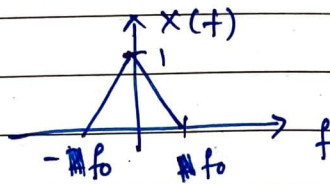


1. $x(t)$:



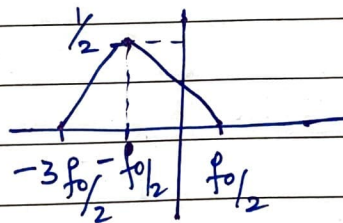
$S_2(t) \leftrightarrow S_2(f)$: $x(t)$ Shifted to right & left by $f_0/2$ ($\because$ of modulation/multiplier) ($\cos \pi f_0 t$)

$S_2(f)$:

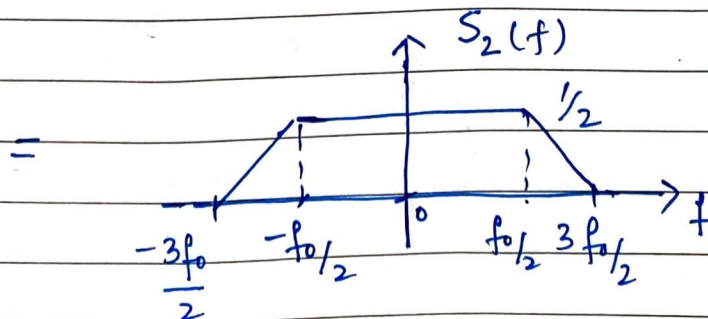
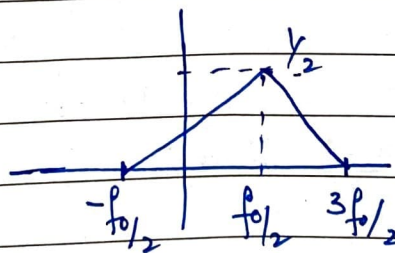


$$S_2(f) = \frac{1}{2} \left(x\left(f - \frac{f_0}{2}\right) + x\left(f + \frac{f_0}{2}\right) \right)$$

$\frac{1}{2} x\left(f - \frac{f_0}{2}\right)$:

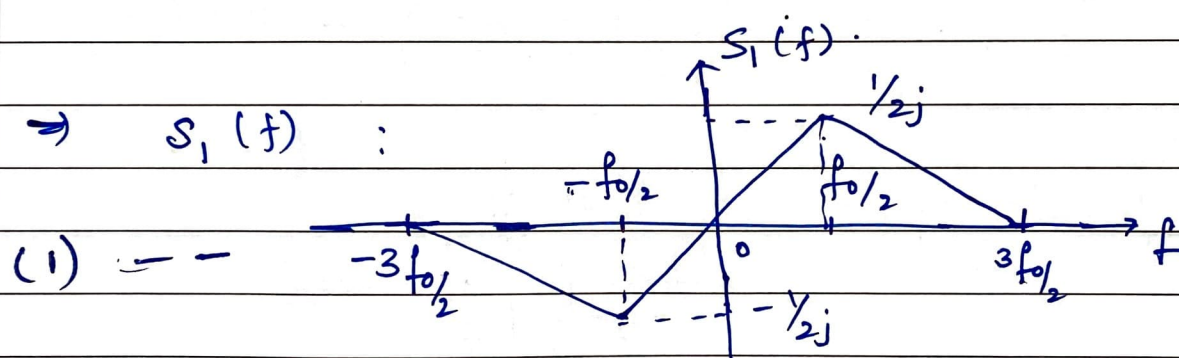
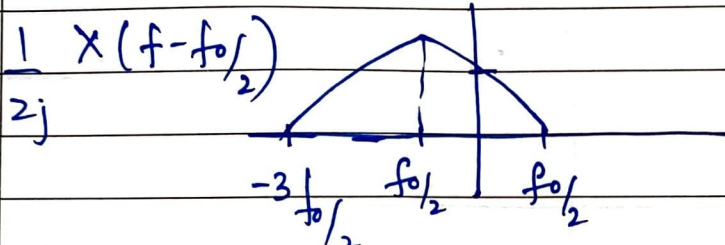
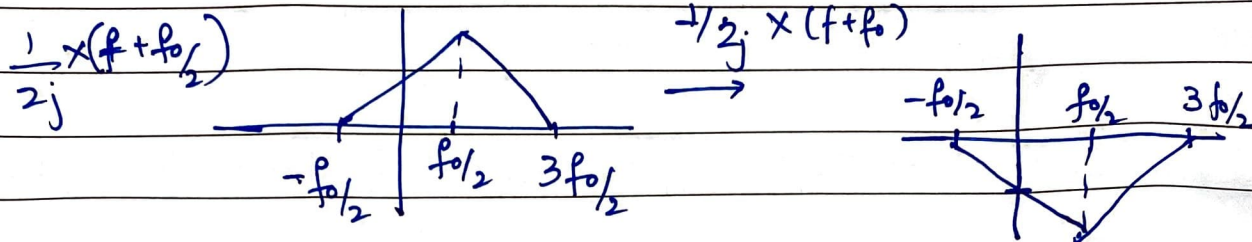


$\frac{1}{2} x\left(f + \frac{f_0}{2}\right)$:

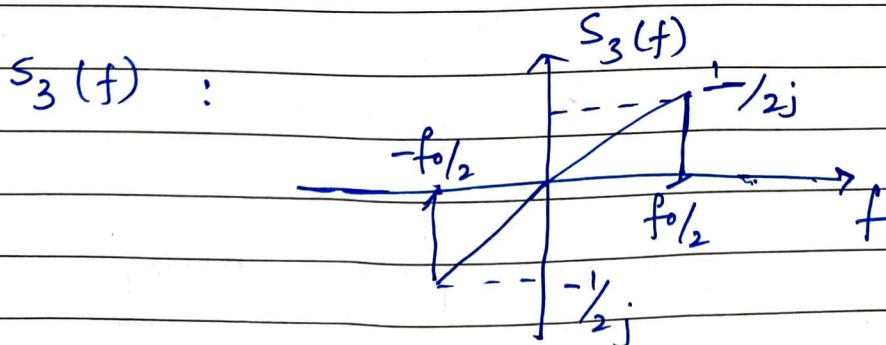


(1)

$$S_1(f) : \frac{1}{2j} \left(x\left(f - \frac{f_0}{2}\right) - x\left(f + \frac{f_0}{2}\right) \right)$$

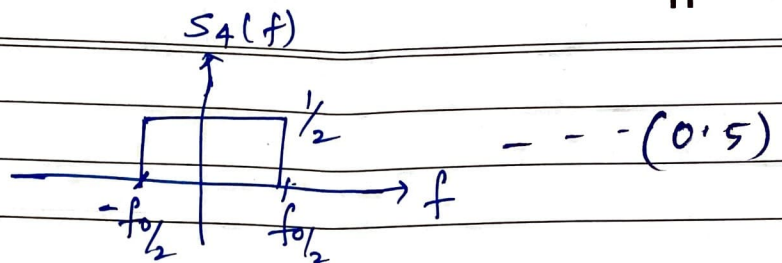


To obtain $S_3(f)$ & $S_4(f)$ we need to pass $S_1(f)$ & $S_2(f)$ through Low pass filter with cutoff $\pm f_0/2$.



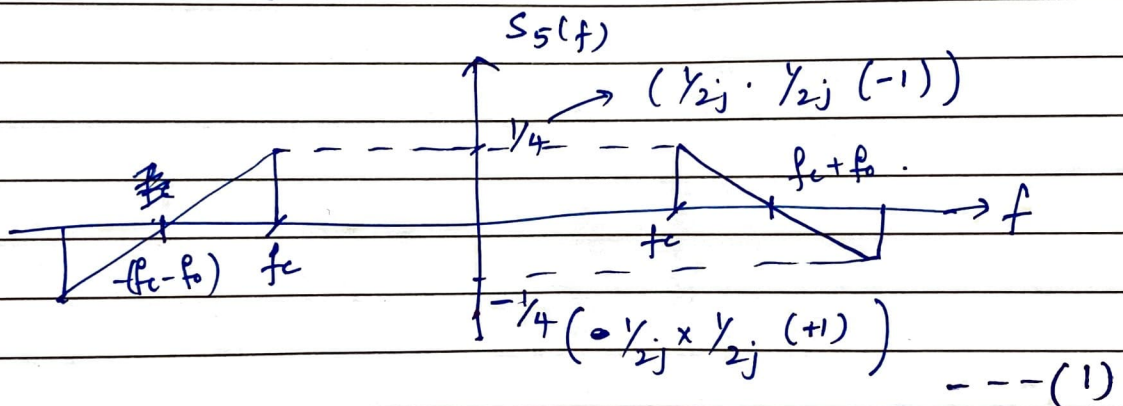
--- (0.5)

$S_4(f)$:



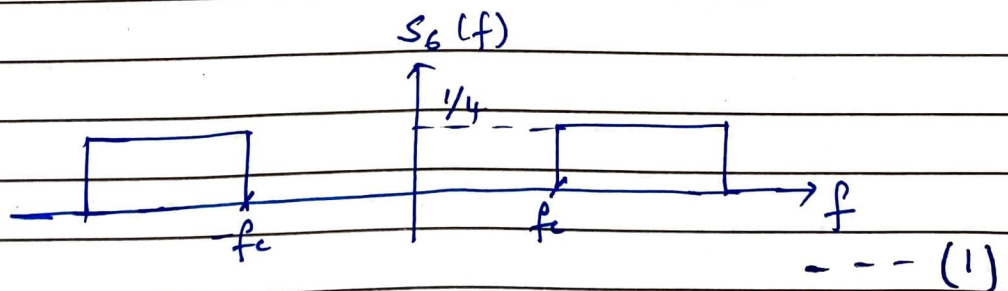
$S_5(f)$: Computing this requires us to modulate the signal again.

$$S_5(f) = \frac{1}{2j} \left[S_3(f - f_c - f_{0/2}) - S_3(f + f_c + f_{0/2}) \right]$$

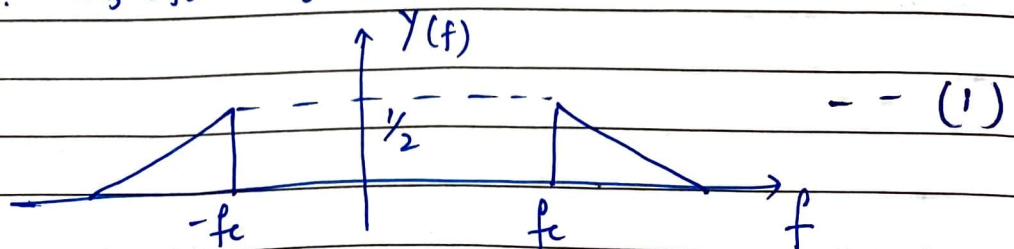


|||y

$S_6(f)$:



$Y(f) : S_5(f) + S_6(f)$.



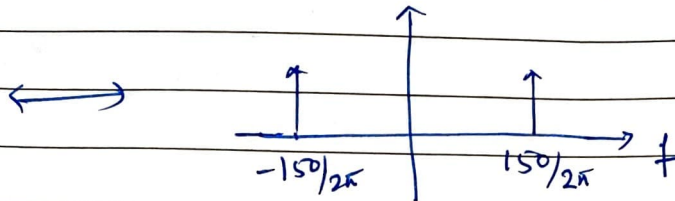
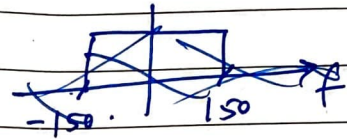
$\rightarrow Y(f)$ is a SSB modulated w/f of $X(f)$ --- (1)

2.(a) Sampling rates of signals:

A. $\sin(300t)$

$= x(t) \Leftrightarrow X(f)$

$|X(f)|$



$\Rightarrow$ Nyquist rate $> \frac{300}{2\pi}$ or $\frac{150}{\pi} \text{ Hz}$ --- (1)

B. $\sin^2(200t) = 1 - 2\cos 400t$

If $x(t) = \cos 400t$, max freq component is $\frac{400}{2\pi} \text{ Hz}$

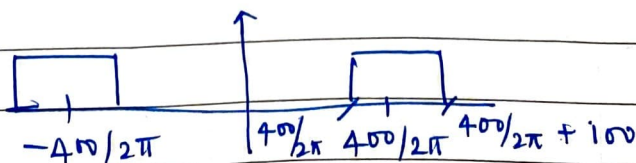
$\Rightarrow$ Nyquist rate $> \frac{400}{\pi} \text{ Hz}$ --- (1)

C. $\cos(300t) + \sin(200t)$

Max frequency component is $\frac{300}{2\pi} \text{ Hz}$

$\Rightarrow$ Nyquist rate $> \frac{300}{\pi}$ --- (1)

d. $m(t) \cos 400t$ max freq component in $m(t)$ is 100 Hz



In modulated signal : Components until $f_c \pm f_m$.

Bandpass sampling cannot be applied here as there will be aliasing.

$$\rightarrow \text{Nyquist rate} \geq \left(\frac{400 + 100}{2\pi} \right) 2$$

$$= \frac{400}{\pi} + 200. \quad \text{--- (1)}$$

$\Rightarrow$ In ascending order: $A < C < B < D$

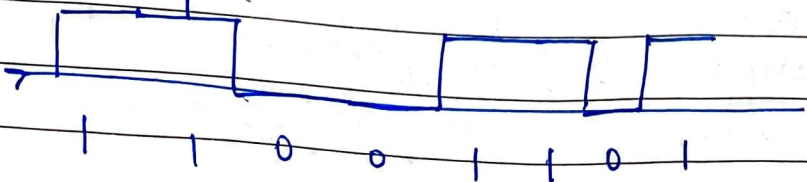
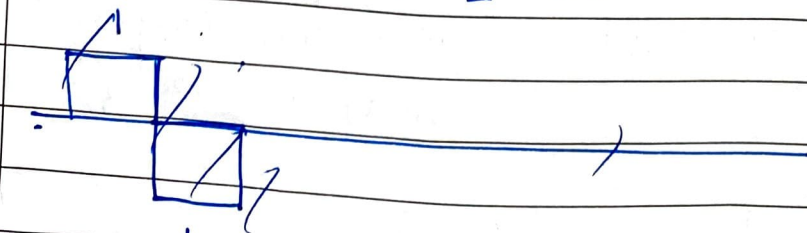
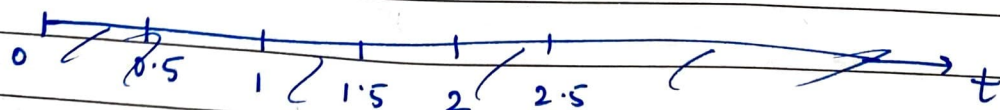
(b) Bandlimited to 5 kHz . Nyquist rate = 10 kHz .

Sampled at 20 kHz (twice Nyq rate) --- (1)

8 bit encoder $\Rightarrow \frac{2^8}{2} = 256$ levels. $2^8 = 256$ levels

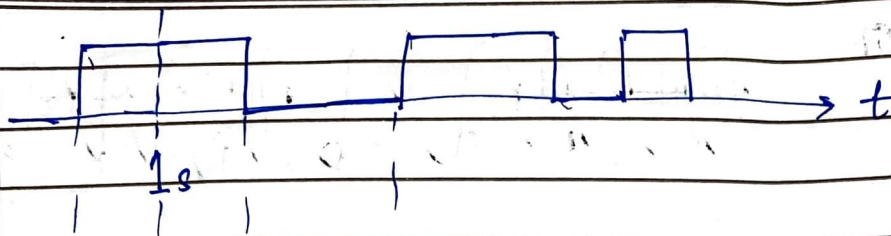
$$20 \text{ kHz} \times 8 = 160 \text{ K samples bits/s} \quad \text{--- (1)}$$

(c) 11001001 $\rightarrow$ 10

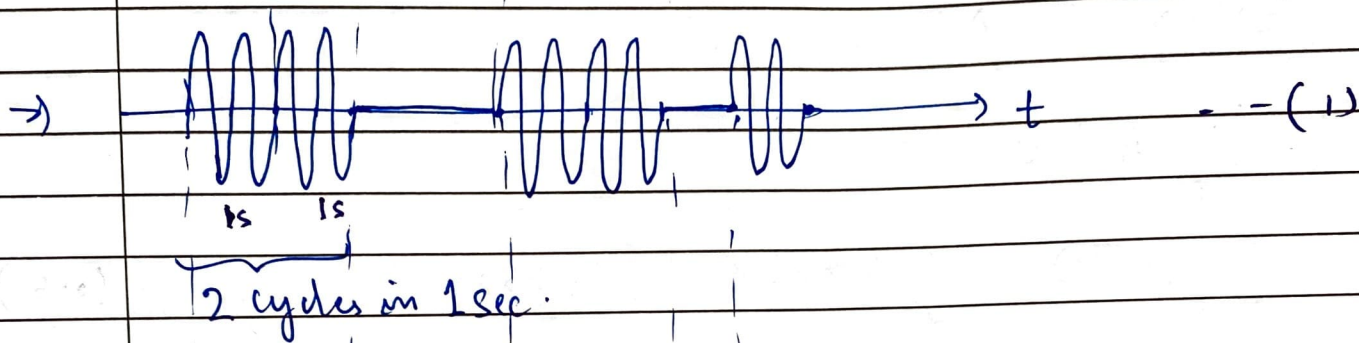


--- (1)

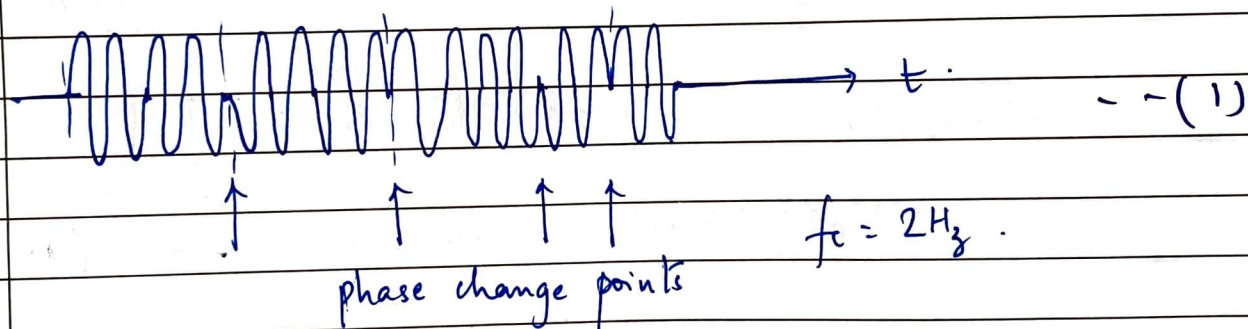
(a)



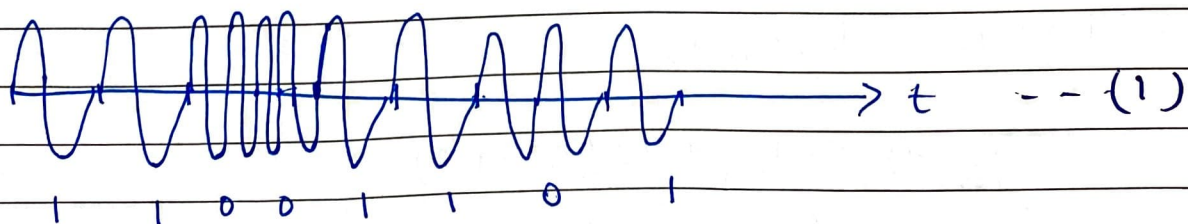
→ 2 cycles with $f_c = 2\text{Hz}$.



(b)

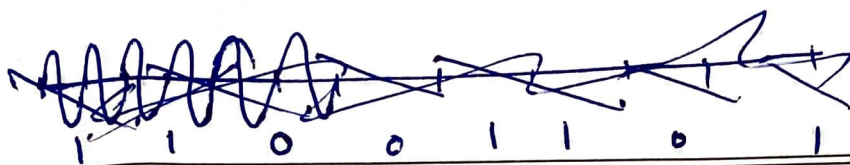


(c) FSK with $f_c = 1\text{Hz}$ for 1 and $f_c = 2\text{Hz}$ (0)
 Students can take $f_c = 1\text{Hz}$ for 0 & " for 1 too.

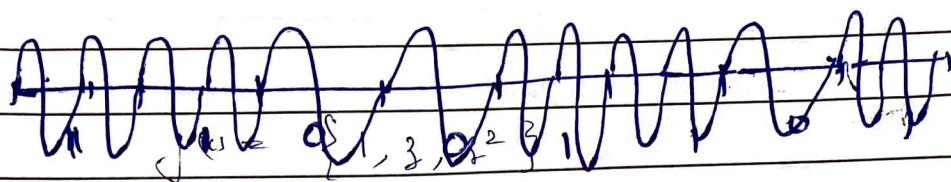


(I have written badly but ensure 0 has 2 cycles while 1 has only 1 cycle.

OR



Rough OR



$$(3) \quad x(t) = A \cos(2\pi t + \theta)$$

$$(a) \quad A \sim U(-1, 1)$$

$$\mu_x(t) = \int A \cos 2\pi t \, E[A \cos 2\pi t] \quad \dots (0.5)$$

$$= E[A] \cos 2\pi t = 0 \quad \because E(A) = 0$$

$$R_x(t_1, t_2) = E[A \cos 2\pi t_1 \cdot A \cos 2\pi t_2]$$

$$= E[A^2] \cos 2\pi t_1 \cos 2\pi t_2 \quad \dots (1.5)$$

$$= \frac{1}{3} \cos 2\pi t_1 \cos 2\pi t_2$$

Although $\mu_x(t)$ is independent of time
 $R_x(t_1, t_2)$ is not a fn of $(t_1 - t_2)$ alone. $\dots (0.5)$
 Hence, process not WSS.

$$(B) \quad \theta \sim U(0, 2\pi)$$

$$\mu_x(t) = E[A \cos(2\pi t + \theta)]$$

$$= \int_{-\infty}^{\infty} A \cos(2\pi t + \theta) p_0(\theta) d\theta \quad \dots (0.5)$$

$$= \int_0^{2\pi} A \cos(2\pi t + \theta) \cdot \frac{1}{2\pi} d\theta = 0 \quad (\text{indep of } t)$$

$$\begin{aligned}
 R_x(t_1, t_2) &= E \left[A \cos(2\pi t_1 + \theta) A \cos(2\pi t_2 + \theta) \right] \\
 &= \frac{A^2}{2\pi} \int_0^{2\pi} \cos(2\pi t_1 + \theta) \cos(2\pi t_2 + \theta) d\theta \\
 &= \frac{A^2}{2\pi} \cdot \frac{1}{2} \int_0^{2\pi} \cos(2\pi(t_1 - t_2)) + \cos(2\pi(t_1 + t_2) + 2\theta) d\theta \\
 &= \frac{A^2}{2\pi} \cdot \frac{1}{2} (2\pi) \cdot \cos(2\pi(t_1 - t_2)) \\
 &= \frac{A^2}{2} \cos(2\pi \tau) \quad \text{where } \tau = t_1 - t_2 \quad \text{--- (1.5)}
 \end{aligned}$$

Hence this process is WSS. --- (0.5)

4. 1. In coin-tossing we only have two outcomes & hence we assign probab p (H) & $(1-p)$ (T).

$$\text{Entropy} = -(p \log_2 p + (1-p) \log_2 (1-p)).$$

$$\begin{aligned}
 \text{If } p=0.5 &: -((0.5) \log_2 (1/2) + 1/2 \log_2 (1/2)) \\
 &= -\log_2 (1/2) = 1 \quad \text{--- (1)}
 \end{aligned}$$

$$\text{If } p=0.7: -(0.7 \log_2 (0.7) + 0.3 \log_2 (0.3))$$

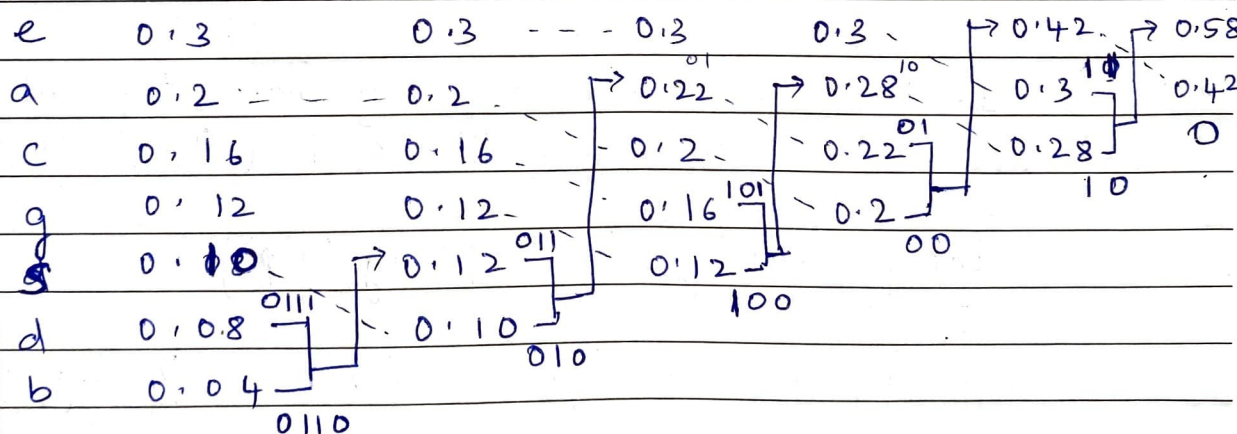
This has lower entropy. --- (0.5)

More information when entropy is high, i.e., for $p=0.5$.
--- (0.5)

2. Huffman encoding:

<u>50</u> total. First find probab of each:		$\frac{8}{50} = \frac{40}{25}$
		0.016
a	0.2	$\frac{2}{50} = \frac{1}{25}$
c	0.16	0.004
b	0.04	$\frac{16}{50} = \frac{8}{25}$
e	0.3	---
d	0.08	(2)
g	0.12	$\frac{6}{50} = \frac{3}{25}$
s	0.1	0.02
	<u>1.0</u>	$\frac{8}{50}$

In descending order:



e 1.
 a 00
 c 101
 g 100
 s 010
 d 0111
 b 0110

* Answer can be different in 2 ways. Check carefully.
 → first step last 2 combined gives 0.12. If student has entered this above g (0.12).
 → At the end, if they start with 0 for 0.58 & 1 for 0.42 & so on.